

FASTTYPINGLAB.COM — COMPLETE AI MULTILINGUAL TYPING PLATFORM MASTER PLAN

Project Overview

FastTypingLab.com will become an AI-powered multilingual typing learning ecosystem focused on:

- English typing
 - Hindi Unicode typing
 - Hindi Kruti Dev typing
 - AI-powered adaptive learning
 - Multiplayer typing battles
 - Govt exam preparation
 - Typing games
 - SEO traffic growth
 - Mobile-first experience
 - Social typing community
-

Technology Stack

Frontend

- Next.js 15+
- TypeScript
- Tailwind CSS
- shadcn/ui
- Framer Motion

Backend

- Node.js
- Supabase
- Supabase Edge Functions

Database

- PostgreSQL via Supabase

AI Integration

- OpenAI API

- Gemini API
- Local analytics engine

Realtime Features

- Supabase Realtime
- WebSockets

Games

- Phaser.js

Charts & Analytics

- Recharts
- Chart.js

Authentication

- Supabase Auth

Hosting

- Vercel

Monitoring

- Sentry
- PostHog

SEO

- SSR + ISR
- Dynamic metadata
- Structured data

Environment Variables

Create `.env.local`

```
OPENAI_API_KEY=  
GEMINI_API_KEY=
```

```
NEXT_PUBLIC_SUPABASE_URL=  
NEXT_PUBLIC_SUPABASE_ANON_KEY=  
SUPABASE_SERVICE_ROLE_KEY=  
  
GOOGLE_ADSENSE_ID=  
  
NEXT_PUBLIC_SITE_URL=https://fasttypinglab.com
```

Core Features

1. Multilingual Typing Engine

Supported Languages

Initial

- English
- Hindi Unicode
- Hindi Kruti Dev

Future Expansion

- Marathi
 - Bengali
 - Punjabi
 - Gujarati
 - Tamil
 - Telugu
-

2. Supported Keyboard Layouts

Hindi Layouts

- Inscript
- Kruti Dev
- Remington Gail
- Phonetic Hindi

English Layouts

- QWERTY

- DVORAK (future)
 - Colemak (future)
-

3. Dynamic Keyboard Layout System

Store layouts in JSON format.

Example:

```
{
  "layout": "inscript",
  "language": "hindi",
  "keys": [
    {
      "code": "KeyK",
      "normal": "क",
      "shift": "ख"
    }
  ]
}
```

Benefits:

- Easy future language expansion
 - Dynamic rendering
 - Layout switching
 - AI layout detection
-

4. Hindi Unicode Typing System

Features

- Real Unicode support
- Browser compatibility
- Govt exam compatibility
- Real-time rendering
- Keyboard visualization
- Finger highlighting

Fonts

- Noto Sans Devanagari
 - Mukta
 - Hind
 - Mangal
-

5. Hindi Kruti Dev Typing System

Features

- Kruti Dev 010 support
- Legacy font rendering
- Real-time conversion
- Keyboard mapping
- Layout visualization

Font

- Kruti Dev 010

Requirements

- Accurate character mapping
 - Real typing calculations
 - Unicode ↔ Kruti conversion support
-

6. AI Auto Layout Detection

AI should detect:

- Inscript
- Kruti Dev
- Remington
- Phonetic

Detection based on:

- Key patterns
- Mistake behavior
- Typing flow
- Character sequences

AI then:

- Suggests best layout
 - Creates optimized exercises
 - Recommends lessons
-

7. Step-wise Learning System

Learning Levels

Beginner

- Finger positioning
- Home row
- Single letters
- Basic combinations

Intermediate

- Words
- Sentences
- Paragraphs
- Speed building

Advanced

- Govt exam practice
 - Timed tests
 - Mixed scripts
 - Long-form typing
-

Lesson Flow

Lesson → Guided Practice → Typing Exercise → AI Analysis → Personalized Drill → Test → Certificate

Lesson Features

- Keyboard visualization
- Finger placement guide

- Animated typing hints
 - Real-time error highlighting
 - Accuracy tracking
 - AI suggestions
 - Voice guidance
-

8. AI Typing Tutor

AI Providers

Primary

- OpenAI API

Secondary/Fallback

- Gemini API

Local Engine

- Typing analytics engine
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AI Features

Weakness Detection

Track:

- Weak keys
 - Slow fingers
 - Repeated errors
 - Hesitation points
 - Shift key problems
 - Matra mistakes
-

Personalized AI Practice

AI generates:

- Daily exercises
 - Smart drills
 - Speed training
 - Accuracy training
 - Exam preparation sets
-

AI Coaching

Provide:

- Motivational feedback
 - Improvement tips
 - Typing posture guidance
 - Daily goals
 - Progress predictions
-

AI Voice Assistant

Languages:

- Hindi
- English

Use:

- Browser Speech API
- ElevenLabs later

Voice feedback:

- "You are slowing on Shift combinations."
 - "Practice the 'क' key more."
 - "Accuracy improved by 5% today."
-

9. Typing Test System

Supported Modes

- English typing test
 - Hindi Unicode test
 - Hindi Kruti Dev test
-

Test Durations

- 1 minute
 - 2 minute
 - 5 minute
 - 10 minute
 - Custom duration
-

Metrics

Calculate:

- WPM
 - CPM
 - Accuracy
 - Error count
 - Net speed
 - Gross speed
 - Backspace usage
 - Typing consistency
-

Features

- Live timer
 - Progress bar
 - Keyboard heatmap
 - Real-time analytics
 - Mistake highlighting
 - Paragraph scrolling
 - Resume support
-

Typing Content Database

Use REAL data only.

Content Sources:

- Hindi news paragraphs
- Govt exam patterns
- Educational passages
- Dynamic AI-generated text

Avoid:

- Dummy paragraphs
 - Fake analytics
 - Simulated scores
-

10. Competitive Exam Module

Supported Exams

- SSC
 - UP Police
 - Court Typing
 - Clerk Exams
 - Hindi Typing Exams
-

Features

- Exact exam simulation
 - Real timer
 - Exam environment
 - Printable result
 - AI mistake analysis
 - Performance ranking
-

11. Multiplayer Typing Battles

Modes

- 1v1 battle
 - Ranked mode
 - Public rooms
 - Private rooms
 - Tournament mode
-

Realtime Features

- Live race tracking
 - Opponent progress
 - Speed graph
 - Live rank updates
-

Future Features

- Voice chat
 - Team battles
 - Clan system
-

12. Fun Zone — Typing Games

Use Phaser.js

Games

English + Hindi Modes

- Word Rain
 - Zombie Typing
 - Space Shooter
 - Speed Racer
 - Falling Letters
 - Multiplayer Arena
-

Features

- XP rewards
 - Combo system
 - Daily challenges
 - Seasonal events
 - Achievements
-

13. Reward Economy

Users earn:

- XP
- Coins
- Badges
- Streak rewards

Coins unlock:

- Themes
 - Keyboard skins
 - Sounds
 - Effects
 - Profile customization
-

Daily Reward System

- Daily login rewards
 - Practice streak rewards
 - Multiplayer rewards
 - Achievement rewards
-

14. Social Platform Features

User Profiles

Include:

- Avatar

- Bio
 - Typing stats
 - Achievements
 - Certificates
 - Rank
-

Social Features

- Follow users
 - Add friends
 - Challenge friends
 - Share achievements
 - Public activity feed
-

Leaderboards

- Global
 - Country
 - State
 - Friends
 - Weekly
 - Monthly
 - All-time
-

15. Certificate System

Generate downloadable certificates.

Include:

- Name
- WPM
- Accuracy
- Language
- Date
- QR verification

Use:

- PDF generation
-

16. YouTube-style Learning Hub

Add:

- Video tutorials
- Hindi typing courses
- Govt exam guides
- Typing strategies

Features:

- Playlists
- Watch history
- Progress tracking
- Notes system

SEO benefit: Huge organic traffic growth.

17. SEO Optimization

Main SEO Keywords

English

- typing speed test
- typing tutor online
- AI typing tutor

Hindi

- Hindi typing test
 - Kruti Dev typing
 - Hindi typing practice
 - Learn Hindi typing
-

SEO Pages

Create pages:

- /hindi-typing-test
- /kruti-dev-typing

- /learn-hindi-typing
 - /typing-games
 - /typing-certificates
 - /competitive-exam-typing
-

Technical SEO

Implement:

- SSR rendering
 - ISR
 - Dynamic metadata
 - Schema markup
 - Sitemap.xml
 - robots.txt
 - Canonical URLs
 - OpenGraph tags
-

Content SEO

Create:

- Blogs
- Tutorials
- Keyboard guides
- Exam preparation articles

Goal: 1000+ SEO pages.

18. Mobile-first Experience

Requirements

- Fully responsive UI
 - PWA support
 - Offline typing practice
 - Push notifications
 - Installable web app
-

Future Apps

- Android app
- iOS app

Recommended:

- React Native OR
 - Flutter
-

19. Admin Dashboard

Features:

- User management
 - Lesson management
 - Analytics
 - AI insights
 - SEO management
 - Ads management
 - Leaderboard moderation
-

20. Institution Dashboard

Institutions can:

- Add students
- Assign tests
- Monitor progress
- Download reports
- Manage batches

Revenue opportunity: B2B subscriptions.

21. Real Data Audit System

Audit all existing features.

Check:

- Dummy leaderboards
- Fake analytics
- Placeholder APIs
- Static paragraphs
- Hardcoded results

Replace with:

- Supabase realtime data
 - Real calculations
 - Live analytics
-

22. Ads Monetization

Use:

- Google AdSense

Placement:

- Blog pages
- Results pages
- Between lessons

Avoid:

- Interrupting typing experience

Future:

- Premium subscription
 - Ad-free plans
-

23. Theme System

Default:

- Light mode

Features:

- Dark mode toggle

- Theme persistence
 - Custom themes
-

24. Security

Implement:

- Rate limiting
 - API validation
 - Anti-cheat system
 - Bot prevention
 - Secure authentication
 - SQL injection prevention
-

25. Anti-cheat System

Detect:

- Copy-paste cheating
 - Unrealistic speeds
 - Automated typing
 - Suspicious behavior
-

26. Analytics Dashboard

Track:

- WPM trends
- Accuracy trends
- Weak keys
- Heatmaps
- Typing consistency
- Practice duration

AI predicts:

- Future WPM
 - Exam readiness
 - Improvement timeline
-

27. Database Structure

Core Tables

users typing_sessions typing_errors typing_heatmaps lessons exercises ai_analysis leaderboards
achievements rewards certificates game_scores multiplayer_rooms followers notifications blog_posts
institutions

28. Suggested Folder Structure

```
/app
  /typing-test
  /learn
  /games
  /multiplayer
  /dashboard
  /blog

/components
  /keyboard
  /typing
  /games
  /ai
  /charts

/lib
  /ai
  /supabase
  /analytics
  /layouts

/data
  /layouts
  /paragraphs
```

29. Recommended Libraries

UI

- shadcn/ui

- Framer Motion

Forms

- React Hook Form
- Zod

State

- Zustand

Charts

- Recharts

Games

- Phaser.js

Analytics

- PostHog
- Plausible

Monitoring

- Sentry
-

30. Development Roadmap

Phase 1

- Hindi typing engine
- Unicode + Kruti Dev
- Learning system
- Real typing tests

Phase 2

- AI tutor
- Voice feedback
- Analytics
- Personalized practice

Phase 3

- Multiplayer battles
- Rewards
- Certificates
- Social features

Phase 4

- Typing games
- SEO expansion
- Blog system

Phase 5

- Mobile apps
 - Regional languages
 - Institution platform
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Final Goal

FastTypingLab.com should become:

- India's #1 typing platform
- AI-powered typing ecosystem
- Govt typing exam hub
- Multiplayer typing community
- Multilingual typing school
- Gamified learning platform

Focus on:

- Real data
- Real analytics
- AI personalization
- SEO scalability
- Excellent UX
- Mobile-first growth